CS319: Scientific Computing

Week 8: Quadrature in 2D; Intro to Classes

Dr Niall Madden

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Slides and examples: https://www.niallmadden.ie/2324-CS319

Outline

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Quadrature 2D

(These slides we part of Week 7, but I didn't get to them in class).

For the last time (in lectures) we'll look at **numerical integration**, this time of two dimensional functions.

That is, our goal is to estimate

$$\int_{a_1}^{b_1} \int_{a_2}^{b_2} f(x_1, x_2) dx_1 dx_2.$$

When we implement an algorithm for this, we will set

- `x1` and `x2` to be vectors of (one-dimensional) quadrature of $N + 1$ points.
- `y` to be a **two-dimensional** array of $(N + 1)^2$ quadrature values. That is, we will set
  `y[i][j] = f(x1[i], x2[j]);`

Derivation

Implementation

We'll implement this for estimating $\int_0^1 \int_0^1 e^{x_1+x_2} dx_1 dx_2$, with N quadrature points in each direction.

00Trap2D.cpp preamble

```
double f(double x1, double x2) {  return(exp(x1+x2)); }
double ans_true = pow(exp(1.0)-1.0,2); // true value

double Trap2D(double *x1, double *x2,
              double **y, unsigned int N);
```

00Trap2D.cpp main()

```
int main(void )
{
  unsigned N = pow(2,4); // Number of points in each direction
  double a1=0.0, b1=1.0, a2=0.0, b2=1.0; // limits of int
  double h1, h2; // step-size in x1 and x2
  double *x1, *x2, **y; // quadrature points and values.

  x1 = new double[N+1];
  x2 = new double[N+1];

  h1 = (b1-a1)/double(N);
  h2 = (b2-a2)/double(N);
  for(unsigned i = 0; i < N+1; i++)
  {
    x1[i] = a1+i*h1;
    x2[i] = a2+i*h2;
  }
```

00Trap2D.cpp main() continued

```
  y = new double * [N+1];
  for(unsigned i = 0; i < N+1; i++)
    y[i] = new double[N+1];

  for (unsigned i=0; i<N+1; i++)
    for (unsigned j=0; j<N+1; j++)
      y[i][j] = f(x1[i], x2[j]);

  double est1 = Trap2D(x1, x2, y, N);
  double error1 = fabs(ans_true - est1);

  std::cout << "N=" <<  N << " | est=" << est1
            << " | error = " << error1 << std::endl;
```

00Trap2D.cpp Trap2D()

```
double Trap2D(double *x1, double *x2, double **y,
              unsigned N)
{
  double Q, h1 = (x1[N]-x1[0])/double(N),
    h2 = (x2[N]-x2[0])/double(N);

  Q = 0.25*(f(x1[0],x2[0]) + f(x1[N],x2[0])   // 4 corners
            + f(x1[0],x2[N]) + f(x1[N],x2[N]));

  for (unsigned k=1; k<N; k++)   // 4 edges (not including corners)
    Q += 0.5*(f(x1[k],x2[0]) + f(x1[k],x2[N])
              + f(x1[0],x2[k]) + f(x1[N],x2[k]));

  for (unsigned i=1; i<N; i++) // All the points in the interior
    for (unsigned j=1; j<N; j++)
      Q += f(x1[i],x2[j]);

  Q *= h1*h2;
  return(Q);
```

Lab 6 preview

- Implement Simpson's Rule in 1D and 2D;
- Verify convergence using Python/NumPy/Jupyter.
- Compare with Monte Carlo(?)

Encapsulation

Encapsulation

Idea: create a single entity in a program that combines data with the program code (i.e., functions) that manipulate that data.
In C++, a description/definition of such entities is called a **class**, and an instance of such an entity is called an **object**.

That is, like a variable is a single instance for a `float` (for example), then an object is a single instance of a `class`.

A class should be thought of as an **Abstract Data Type** (ADT): a specialised type of variable that the user can define.

There are many important examples of "built-in" C++ classes, such as `string`, and objects, such as `cin` and `cout`. But we'll leave those until later, and first study how to make our own.

Encapsulation

The next bit is really important: not just to C++, but for writing robust scientific computing code.

Within an object, code and data may be either

- **Private**: accessible only to another part of that object, or
- **Public**: other parts of the program can access it even though it belongs to a particular object. The public parts of an object provide an **interface** to the object for other parts of the program.

It is referred to a **"data hiding"**, an important concept in software design.

In C++, *encapsulation* is implemented using the `class` keyword. The example we'll consider is a **stack** – a *LIFO* (Last In First Out) queue.

. . . . . .

There is already a C++ implementation of a `stack`*. It is part of the* **Standard Template Library (STL)***. We reinvent the wheel here only because it is a nice example that includes most of the key concepts associated with classes in C++. We will study the STL later in CS319.*

The name of our class will be `MyStack`. It will permit two primary operations:

- an item may be added to the top of the stack: `push()` ;
- an item may be removed from the top of the stack: `pop()`.

These then are our interfaces to the stack. Hence these will be **public**.

For the stack itself, the following must be maintained:

- an array containing the items in the contents;
- a counter/index to the top of the stack.

These are *private* to the class.

We choose this example because it is obvious that

- `push()` and `pop()` are the interfaces to the object–they are declared as *public*;
- the contents of the stack, and the counter of the number of objects in it, need only be visible to the object itself; hence they are private.

In our example there is also a public function to initialise the stack.

The basic syntax for defining a class:

```
class class-name {
private:
    ...     // private functions and variables
public:
    ...     // public functions and variables
};
```

class-name becomes a new object type—one can now declare objects to be of type *class-name*.

This is only a declaration. Therefore,

- functions are not defined, though the prototype is given,
- variables are declared but are not initialised,
- the declaration block is delineated by **{** and **}**, and terminated with a semicolon.

As mentioned our class has two `private` members

- `contents`: a *`char`* array of length `MAX_STACK` the array containing the stacked items.
- `top`: an *`int`* that stores the number of items on the stack.

It has three `public` member functions:

(a) `init()` sets the stack counter to 0. No arguments or return value.

(b) `push()` adds an item to the stack. One argument: the character to be added.

(c) `pop()` takes no argument but returns the removed item.

```
class MyStack {
private:
  char contents[MAX_STACK];
  int top;
public:
  void init(void );
  void push(char c);
  char pop(void );
};
```

To define the functions associated with a particular class we use

1. the name of the class, followed by
2. the *scope resolution operator* `::` , followed by
3. the name of the function.

We now define the three (public) functions: `init()`, `push()` and `pop()`.

The `init()` is required only to set the value of `top` to zero:

```
void MyStack::init(void)
{
  top=0;
}
```

Note that we didn't have to declare the (private) variable `top`.

The `push()` function takes as its only argument a single character. It adds the character to the stack and increments the index to the top of the stack.

```
void MyStack::push(char c) {
  contents[top]=c;
  top++;
}
```

The pop() function doesn't take any arguments (void). It removes the item from the stack by returning the top entry and decrementing top.

```
char MyStack::pop(void){
  top--;
  return(contents[top]);
}
```

The first item in the stack is at position 0, the second is a position 1, the 3rd is at position 2, etc. So when top=n then there are n items in the stack but the top one is actually located in contents[n-1].

Now that our class `MyStack` has been declared, and its functions defined, we can declare objects to be of type `MyStack`, e.g.,

```
MyStack s1, s2;
```

We can refer to the functions `s1.pop()` and `s2.push(c)`, say, because these are public members of the class. We cannot refer to `s1.top` as this variable is private to the class and is hidden from the rest of the program.

. . . . . .

To use the objects, we could have a `main()` function that behaves as follows:

- Declare and initialise a `MyStack` object `s`;
- Push the characters `'C','S','3','1','9'` onto the stack;
- The stack's contents are popped and output to the console using `cout`.

01MyStack.cpp

```
int main(void ) {
  MyStack s;

  s.init();

  s.push('C');
  s.push('S');
  s.push('3');
  s.push('1');
  s.push('9');

  std::cout << "Popping ... " << std::endl;
  std::cout << s.pop() << std::endl;
  std::cout << s.pop() << std::endl;
  std::cout << s.pop() << std::endl;
  std::cout << s.pop() << std::endl;
  return (0);
```

Constructors

Suppose we wanted to change the `MyStack` class so that the user can choose the maximum number of elements on the stack...

In the example above, the function `init()` is used explicitly to initialise the variable `top`. However, there is an initialisation mechanism called a **Constructor** that is built into the concept of a class.

CONSTRUCTOR

A **Constructor** is a public member function of a class

- that shares the same name as the class, and
- is executed whenever a new instance of that class is created.

Constructors

Constructors may contain any code you like; but it is good practice to only use them for initialization.

As an example, we'll change the declaration of the `stack` class as shown here:

```
class MyStack {
public:
  MyStack(void); // Constructor. No return type
  void push(char c);
  char pop(void );
private:
  char contents[MAX_STACK];
  int top;
};
```

Constructors

We then replace the `init()` function with:

```
MyStack::MyStack(void )
{
  top=0;
}
```

Note that the constructor as no explicit return type.

Now whenever an objects of type `MyStack` is created, e.g., with
`MyStack s`,
the function `s.MyStack()` is called automatically – and *`s.top`* is set to zero.

Constructors

We now make the following modifications to the `stack` implementation (for full implementation, see `02MyStackConstructor.cpp`)

```
class MyStack {
private:
  char *contents;
  int top, maxsize;
public:
  MyStack (void);
  MyStack (unsigned int StackSize);
  void push(char c);
  char pop(void );
};
```

Here we have changed `contents` so that it is a pointer.

Constructors

Code for the constructor.

```
MyStack::MyStack(void)
{
  contents = new char [MAX_STACK];
  top=0;
}
```

Destructors

Complementing the idea of a constructor is a **destructor**. This function is called

- for a local object – whenever it goes out of scope,
- for a global object – when the program ends.

The name of the destructor is the same as the class, but preceded by a tilde:

```
class MyStack {
private:
  char *contents;
  int top;
public:
  MyStack(void );
  ~MyStack(void );
  void push(char c);
  char pop();
};
```

Destructors

```
MyStack::~MyStack()
{
  delete [] contents;
}
```

The example we had earlier of a constructor was particularly basic, not least because is its parameter list is `void`. More commonly, one passes arguments to the constructor that can be used, e.g.,

- to set the value of a data member;
- dynamically size an array using `new`.

However, one should still provide a default constructor (i.e., one with no arguments), or one with a default argument list.

[1]

```
class MyStack
{
private:
  char *contents;
  int top;
public:
  MyStack(void);
  MyStack(unsigned  SkSize);
  void push(char c);
  char pop(void );
};
```

```
MyStack::MyStack(void)
{
  top=0;
  contents = new char[MAX_STACK];
}

MyStack::MyStack(unsigned SkSize)
{
  top=0;
  contents = new char[StackSize];
}
```

[1] This is for illustration. Better again: use one constructor, but with a default argument value.